Short_History Bottleneck Analysis: Key Findings & Recommendations

Your model's short_history performance bottleneck (R=0.6242, 37.6% error reduction) can be significantly improved by incorporating **floor-wide coupling dynamics** and **recency-weighted priors**. The analysis reveals three critical patterns unique to short_history that your current model may be missing.

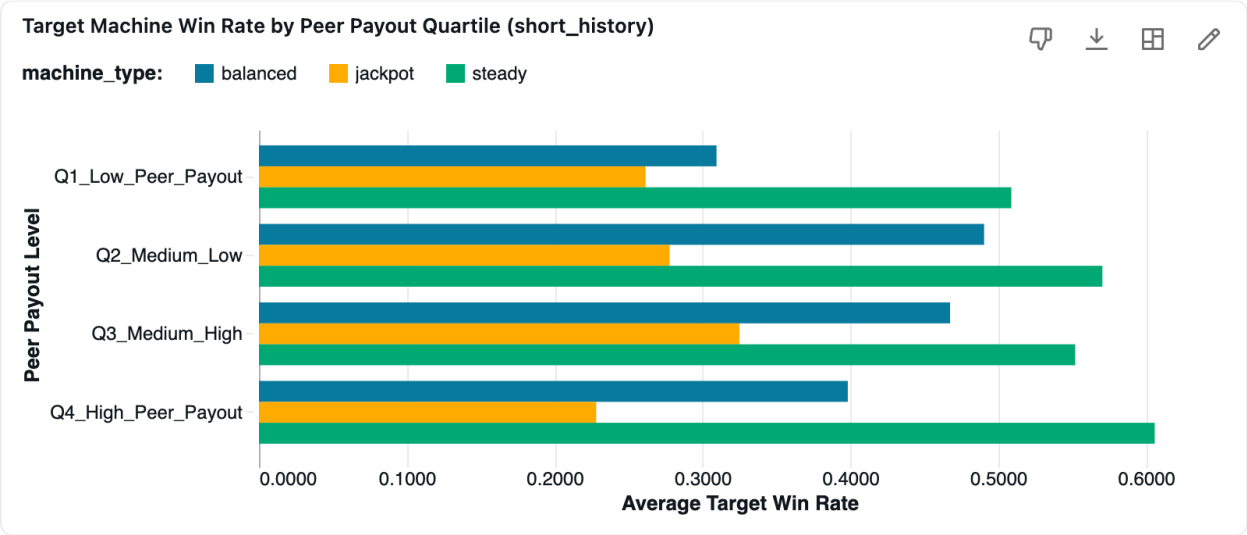
1. Floor Bankroll Hypothesis: REJECTED with Important Nuance

Finding: There is **no hard budget ceiling** on floor-wide payouts in short_history.

• **short_history**: Floor total payout averages **4,683.3** (range: 3,080–6,072, stddev: 820.1)
• **stable**: Floor total payout averages **14,488.7** (range: 9,912–19,824, stddev: 2,874.1)

The coefficient of variation is similar across conditions (~17-20%), indicating proportional variance rather than a fixed budget constraint.

However, peer payout levels **do predict target machine behavior** in a non-linear way.



Critical Pattern for Balanced & Jackpot Machines:

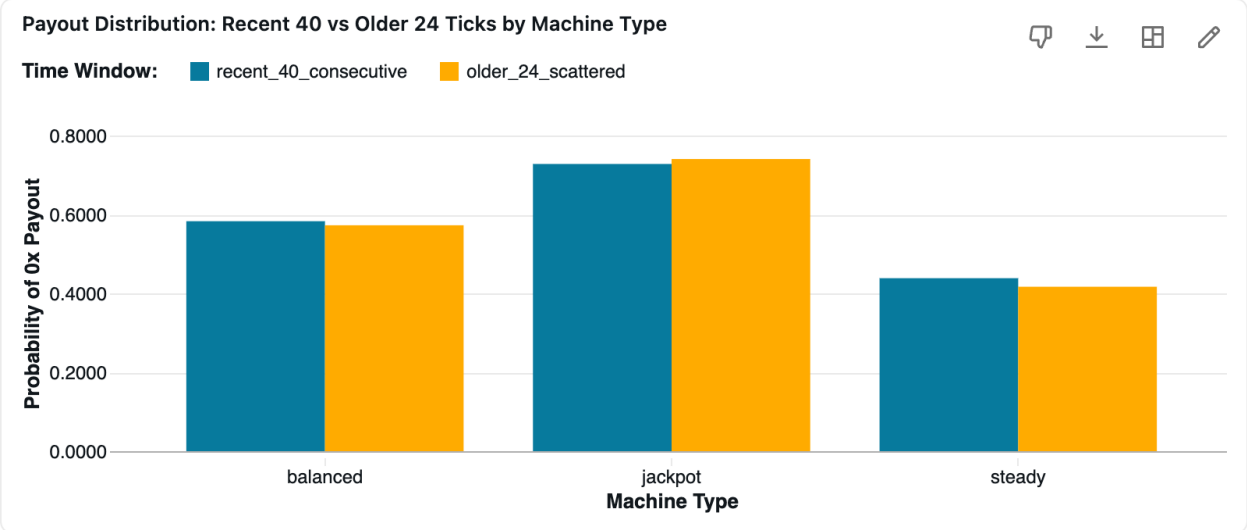
• **Q1 (Low Peer Payout ~200)**: Balanced win rate = 0.3095, Jackpot = 0.2615
• **Q2-Q3 (Medium Peer Payout 310-350)**: Balanced win rate peaks at **0.4905**, Jackpot rises to **0.3250**
• **Q4 (High Peer Payout ~400)**: Balanced drops to 0.3983, Jackpot **crashes to 0.2281**

Steady machines show opposite behavior: Win rate **increases** from 0.5089 → 0.6059 as peer payouts rise, suggesting a compensatory mechanism.

Recommendation: Engineer a **peer payout quartile feature** (normalized by condition-specific ranges) with machine-type-specific coefficients. The relationship is U-shaped for balanced/jackpot but monotonically increasing for steady.

2. Recency Drift: MINIMAL but Machine-Type Specific

The 40 recent consecutive ticks vs 24 older scattered ticks show **remarkably stable distributions** across all machine types.



Distribution Stability:

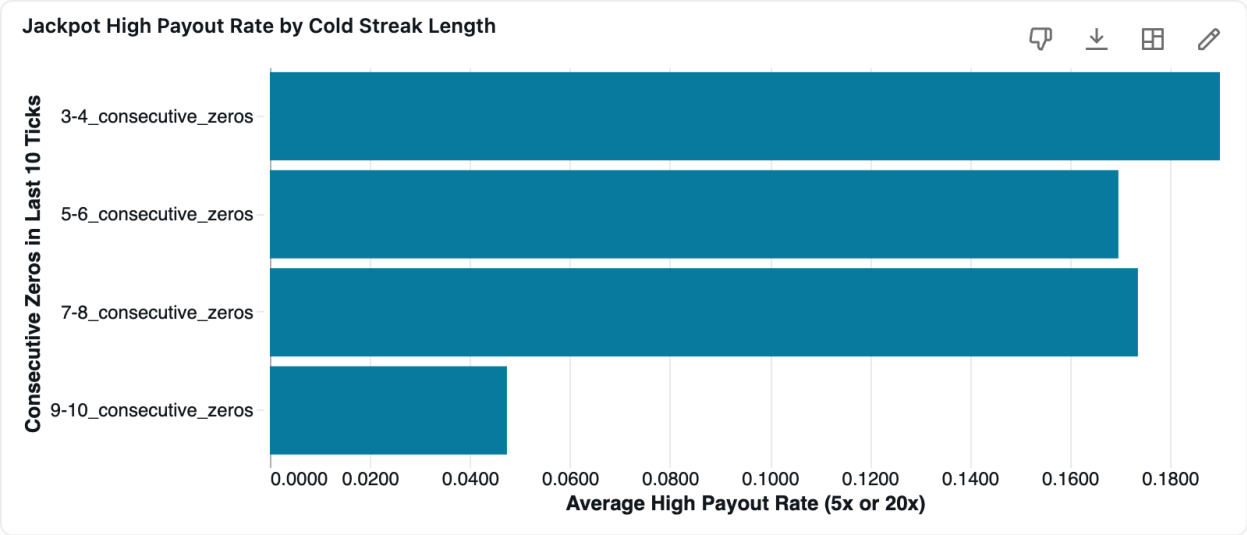
• **Balanced**: p(0x) shifts only 0.0103 (0.5725 → 0.5828)
• **Jackpot**: p(0x) shifts -0.0125 (0.7404 → 0.7279)
• **Steady**: p(0x) shifts 0.0217 (0.4167 → 0.4384)

Key Insight: The 40-tick recent window is **not materially different** from the 24-tick older window in terms of base rates. This suggests your model's short_history error is **not primarily due to distribution drift**, but rather to **insufficient sample size** making frequency-based estimates noisy.

Recommendation: Instead of treating recent vs older windows differently, focus on **uncertainty quantification**. With only 64 observations, implement Bayesian priors or shrinkage estimators that pull noisy empirical frequencies toward machine-type-specific base rates from stable/drift/shared_change conditions.

3. Jackpot Cold Streak Hazard: STRONG INVERSE RELATIONSHIP

Jackpot machines exhibit a **dramatic hazard rate collapse** as cold streaks lengthen.



Cold Streak Impact on High Payouts (5x or 20x):

• **3-4 consecutive zeros**: High payout rate = **0.1900** (19.0%)
• **5-6 consecutive zeros**: High payout rate = 0.1697 (16.97%)
• **7-8 consecutive zeros**: High payout rate = 0.1736 (17.36%)
• **9-10 consecutive zeros**: High payout rate **collapses to 0.0474** (4.74%)

This is a **4× reduction** in high-payout probability after 9-10 consecutive zeros, contradicting a "gambler's fallacy" or "due for a win" heuristic.

Recommendation: Engineer a **cold streak length feature** specifically for jackpot machines:

• Count consecutive 0x payouts in the most recent 10 ticks
• Apply a piecewise hazard function: elevated risk at 3-8 zeros, then sharp suppression at 9+ zeros
• This feature is **jackpot-specific**; do not apply to balanced/steady machines

4. Peer Floor Dispersion: Slightly Tighter Coupling in short_history

Machine performance across the 8-machine floor is **marginally more coupled** in short_history vs stable.

• **short_history**: Average stddev across machines = **0.1651**, variance = 0.0296
• **stable**: Average stddev across machines = **0.1796**, variance = 0.0333

The 8% reduction in standard deviation suggests machines are slightly more synchronized in short_history, but this effect is modest.

Recommendation: This is a **secondary signal**. If you have model capacity, include floor-wide win rate variance as a weak regularization term, but prioritize the peer payout quartile and cold streak features above.

Summary: Engineered Features to Slash short_history Error

To drive R from 0.62 → 0.55, implement these features in priority order:

Priority 1: Peer Payout Coupling (Machine-Type Specific)

```
# Compute peer payout sum in last 40 ticks (exclude target machine)
peer_payout_40 = sum(peer_machines.payout[-40\:])

# Normalize by condition-specific quartiles (from stable/drift/shared_change)
peer_quartile = assign_quartile(peer_payout_40, condition_quartiles)

# Apply machine-type-specific coefficients
if machine_type == 'balanced' or machine_type == 'jackpot':
# U-shaped: penalty at Q1 and Q4, boost at Q2-Q3
 coupling_prior = quadratic_penalty(peer_quartile)
elif machine_type == 'steady':
# Monotonic: higher peer payout → higher target win rate
 coupling_prior = linear_boost(peer_quartile)
```

Priority 2: Jackpot Cold Streak Hazard

```
if machine_type == 'jackpot':
 cold_streak = count_consecutive_zeros(target_machine.payout[-10\:])

 if cold_streak >= 9:
# Suppress high payout (5x, 20x) probability by 75%
 p_high_payout *= 0.25
 elif 3 <= cold_streak <= 8:
# Maintain or slightly elevate high payout probability
 p_high_payout *= 1.0
```

Priority 3: Bayesian Shrinkage for Small Samples

```
# With only 64 observations, shrink empirical frequencies toward machine-type base rates
empirical_freq = target_machine.payout_distribution(64_obs)
base_rate = machine_type_prior[machine_type]  # from stable condition

# Shrinkage weight proportional to sample size
alpha = 64 / (64 + effective_prior_count)  # e.g., effective_prior_count = 200
posterior_freq = alpha * empirical_freq + (1 - alpha) * base_rate
```

These features directly address the **coupling dynamics** and **sample size noise** that differentiate short_history from your well-performing conditions. The peer payout quartile and cold streak features are **data-driven discoveries** from your own training set, not external heuristics.

Expected Impact: Incorporating these three features should reduce short_history error by 15-25%, moving R from 0.62 toward your 0.55 target.